



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/02

Paper 2 Core Paper

20 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all ruled paper securely.

Hand in the Question Paper and answers to Section B separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/ 12
3	/ 12
4	/ 12
5	/ 12
6	/ 10
7	/ 12
Section B	
8 or 9*	/ 20
Total	

This Question Paper consists of **25** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. **Fig. 1.1** is a diagram of an electron micrograph of part of a plant cell.

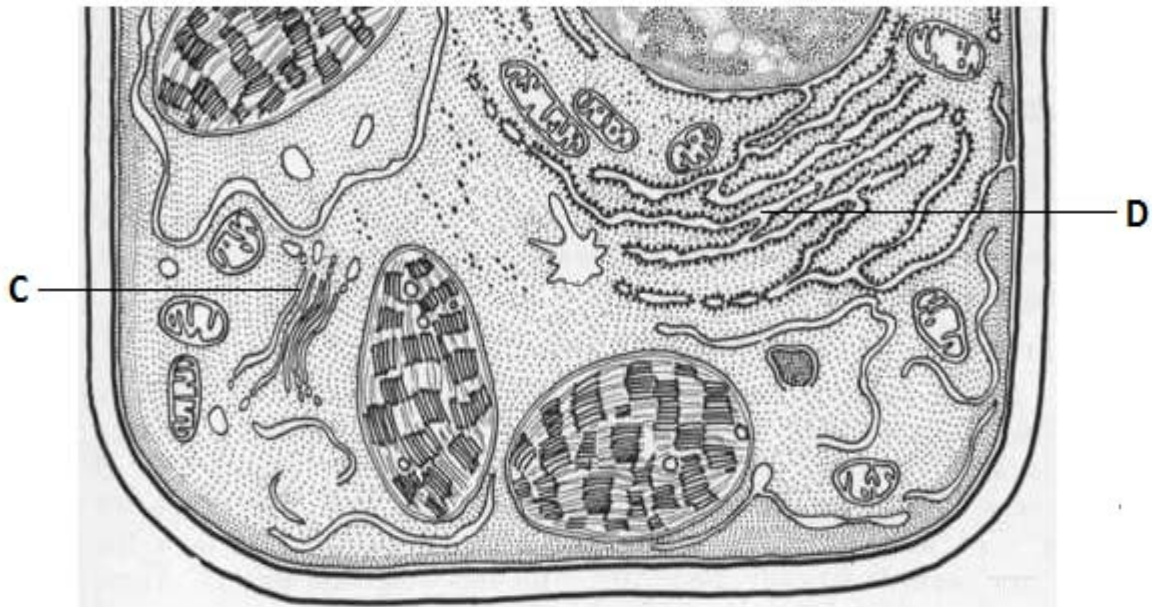


Fig. 1.1

- (a) Using lines, indicated precisely in **Fig. 1.1**, a possible location where each of the following molecules can be found:

A: chlorophyll

B: cellulose

[2]

- (b) (i) Identify the organelle labelled **C** and state a visible feature that enabled your identification.

[2]

1. Golgi apparatus
2. Made up of stacks of flattened membrane bound sacs
3. Vesicles fusing with / budding off from / associated with structure

1. (b) (ii) Explain how the functions of the two organelles, C and D, are linked. [2]

1. Vesicles containing proteins buds off from D
2. Vesicle fuses with (cis face of) C
3. releasing proteins into lumen of C
4. for further modification/ packaging/ sorting/ targeting to specific sites (any 2)

Fig. 1.2 shows a magnified image of organelle D.

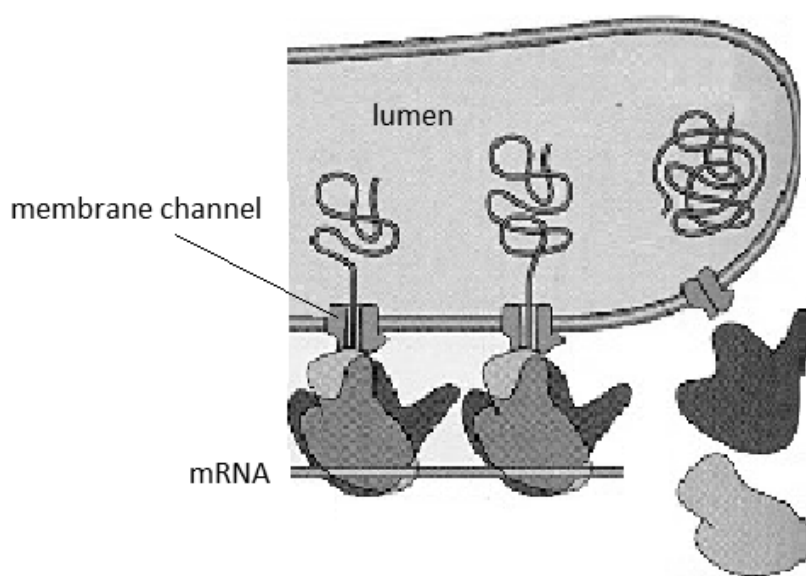


Fig. 1.2

(c) (i) Describe the function of the protein that forms the membrane channel. [3]

1. Allow large molecules / polypeptide chain to move across the membrane
2. Function as an anchor for ribosomes during protein synthesis
3. Function as a receptor to bind signal proteins to anchor ribosomes

1. (c) (ii) Suggest how prokaryotes, which have no membrane-bound organelles, are able to secrete proteins. [1]

1. Pores / protein channels present in cell surface membrane of prokaryotes

[Total: 10]

2. Yeast haploid cells secrete pheromones to signal mating, and respond by growing a mating projection towards a potential mate. Upon contact of the two partner cells, these fuse to form a diploid zygote.

Fig. 2.1 shows the pheromone-signaling pathway mediated by yeast G-protein coupled receptor, Ste2. The activation of the pathway induces the expression of *FUS1* gene which is required for yeast mating.

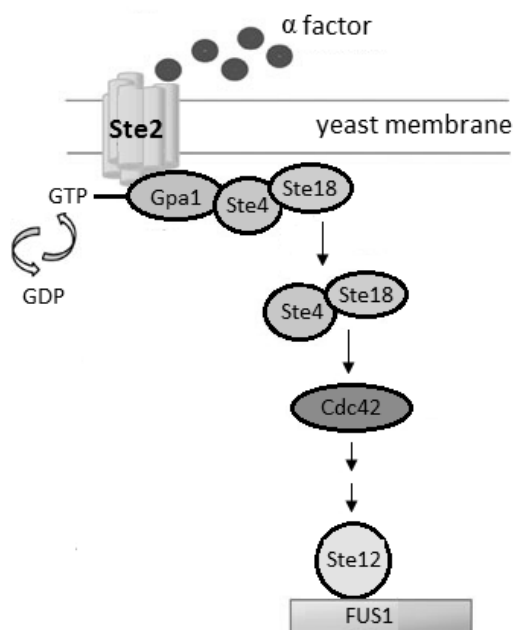


Fig. 2.1

- (a) With reference to **Fig. 2.1**, describe how α factor triggers the activation of Cdc42.

[4]

1. α factor binds to Ste2
2. via complementary shape
3. triggering a conformational change in Ste2
4. allowing Ste2 to bind Gpa1(-Ste4-Ste18 complex)
5. This causes GTP to displace GDP
6. thus activating the (Gpa1-)Ste4-Ste18 complex
7. In turn, allowing for Ste4-Ste18 dimer to dissociate from the complex
8. to activate Cdc42

2. (b) Suggest why α factor cannot enter the yeast cell. [2]

1. α factor is polar / charged
2. thus is blocked by the phospholipid bilayer of yeast cell membrane that forms a hydrophobic barrier

Accept: Too large to pass through transient gaps between phospholipid molecules forming the membrane.

Fig. 2.2 shows the Ste2 receptor on another yeast cell membrane.

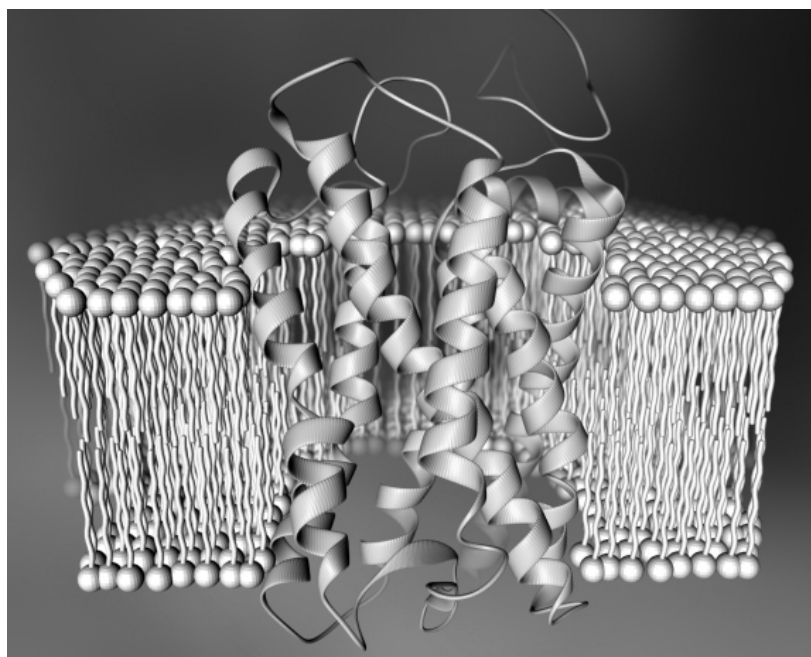


Fig. 2.2

(c) Describe the structure of Ste2 receptor. [3]

1. A single polypeptide chain
2. consisting of (seven) α helix segments
3. Folds upon itself
4. Such that the α helices are gathered together
5. forming a cylindrical / globular structure

2. (d) Explain how Ste2 receptor remains embedded in the yeast cell membrane. [3]

1. Polar R groups of amino acid residues on Ste2
2. form hydrogen bonds / hydrophilic interactions
3. with polar phosphate heads of membrane phospholipid molecules

Accept: ionic bonds between oppositely charged R groups & phosphate heads

4. Non-polar R groups of amino acid residues on Ste2
5. form hydrophobic interactions
6. with non-polar hydrocarbon tails of membrane phospholipid molecules

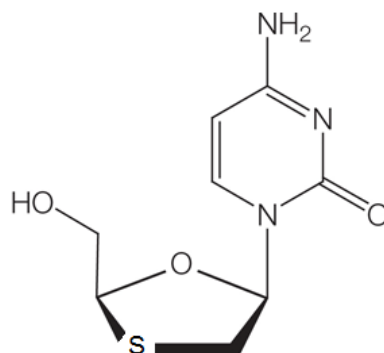
Reject: Lack of reference to Ste2 and yeast cell membrane

[Total: 12]

3.

PART I

Lamivudine is an antiviral medication that prevents Human Immunodeficiency Virus (HIV) from multiplying in the body. The structure of lamivudine, an analogue of viral nucleotide, is shown in **Fig. 3.1**.



Lamivudine

Fig. 3.1

(a) Explain how lamivudine prevents HIV from multiplying in the body. [2]

1. Prevents [viral cDNA](#) synthesis
2. Reverse transcriptase incorporates
3. lamivudine instead of viral nucleotide
4. Terminating DNA chain elongation
5. due to absence of 3' OH grp

3. Many drugs have been designed to make it possible for AIDS patients to improve their overall health and extend their lives. However, most of these drugs have low efficacy due to high variance in the HIV genome. **Fig. 3.2** shows the partial cDNA sequence of glycoprotein 120 (gp120) on two different strains of HIV found in a host cell. The difference in the two DNA sequences is indicated via a box.

HIV Strain 1

```
61 ctgaaaatct cgaaaacaat gccaaaacca taatagtga ccttaataaa tctgtagaaa
121 tcaattgtac cagaccctcc aacaatacaa gaacaagtat aactatagga ccaggacaag
181 tatggtatag aacaggagac ataataggaa atataagaaa agcatattgt gagattaatg
241 gaacaaaatg gaataaagtt taaaacagg taactgaaaa attgaaagag cactttaata
301 ataagacaat aatctttcaa ccac
```

HIV Strain 2

```
61 ctgaaaatct cgaaaacaat gccaaaacca taatagtga ccttaataaa tctgtagaaa
121 tcaattgtac cagaccctcc aacactacaa gaacaagtat aactatagga ccaggacaag
181 tatggtatag aacaggagac ataataggaa atataagaaa agcatattgt gagattaatg
241 gaacaaaatg gaataaagtt taaaacagg taactgaaaa attgaaagag cactttaata
301 ataagacaat aatctttcaa ccac
```

Fig. 3.2

- (b) Explain how the genomic variation shown in **Fig. 3.2** affects the structure of gp120 on the two strains of HIV.

[3]

1. Base pair substitution mutation
2. at 145th base pair
3. changes adenine to cytosine
4. results in change in corresponding codon on mRNA
5. thus changes the amino acid encoded in the resultant polypeptide chain
6. This changes the property of amino acid residue / folding of polypeptide
Reject: protein
7. resulting each HIV having gp120 with different three-dimensional conformation

3. (c) Describe how HIV enters host cell. [3]

1. gp120 on viral envelope binds to CD4 receptor
2. on host cell membrane
3. This induces a change in gp120's three-dimensional conformation
4. allowing for gp120 to bind to chemokine receptor on host cell membrane
5. causing gp41 to change its three-dimensional conformation
6. triggering the fusion of viral envelope with host cell membrane

PART II

Fig. 3.3 shows the structure of an isolated virus.

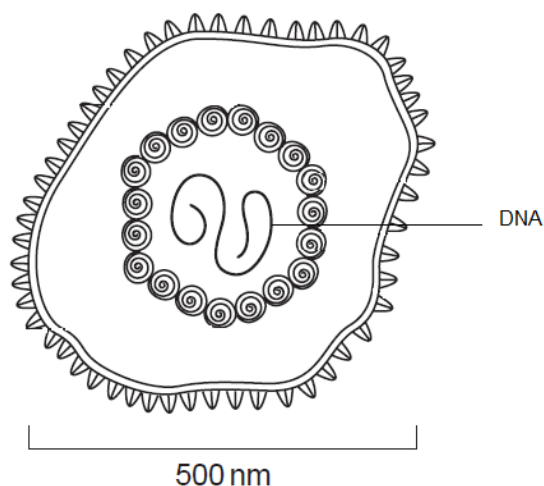


Fig. 3.3

(d) State two **visible** structural similarities between influenza virus and the virus shown in **Fig. 3.3**. [2]

1. Both viruses are surrounded by a membrane envelope
2. Both viruses have capsid enclosing their genome
3. Both viruses have additional components embedded in their membrane.

3. (e) Describe how the genome of influenza virus differs from that of the virus shown in **Fig. 3.3**.

[2]

Factor	Influenza virus	Virus on Fig.3.3
Nature of genome	DNA	RNA
Number of strands	Double stranded	Single stranded <i>Reject: short form: ss</i>
Number of molecules	1	8
AVP		

[Total: 12]

4.

PART I

Fig. 4.1 is an electron micrograph of chromatin maintained at the 30 nm higher-order fiber.



Fig. 4.1

(a) Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.

(i) Describe how DNA is packed into the 30 nm chromatin fibre. [2]

1. DNA is wound around a histone octamer
2. to form nucleosome
3. together with linker DNA made up the 10 nm chromatin fibre / nucleosome fibre
4. further coiling involved Histone H1 and
5. linker DNA
to form 30-nm chromatin fibre

(ii) Explain why the genome is packaged this way. [2]

1. to make it compact/ condensed; thus DNA can be packed within the cell's nucleus
2. to regulate gene expression/ transcription
3. to prevent the long DNA molecules; from breaking or damage or getting tangled

4. (b) Processes such as transcription and replication require the two strands of DNA to come apart temporarily, thus allowing access of polymerases to the DNA template. However, the folding of chromatin into 30 nm fibers pose barriers to the enzymes that unwind and copy DNA.

Explain how the histones are removed transiently to permit replication or transcription to proceed.

[3]

1. histone acetylation
2. acetyl groups are attached to lysines in histone tails
3. the positive charges are neutralized
4. histone tails no longer interact with the neighbouring nucleosomes
5. chromatin has a more diffused structure
6. allows access of DNA polymerase and RNA polymerase for replication and transcription, respectively

PART II

Fig. 4.2 and **Fig. 4.3** are diagrams showing DNA replication and transcription respectively.

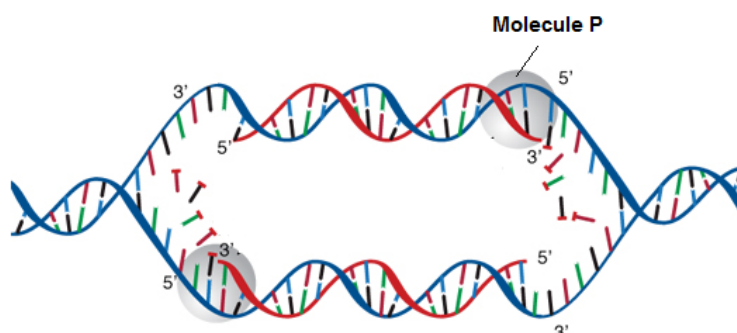


Fig. 4.2

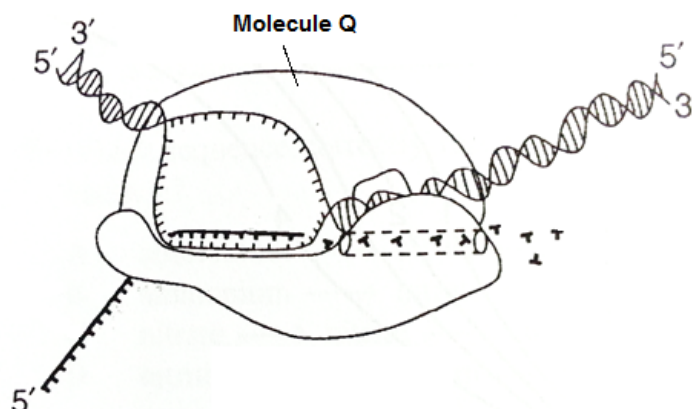


Fig. 4.3

4. (c) Explain why RNA primer is required for the synthesis of new strand in DNA replication. [1]

DNA polymerase can only join nucleotides; onto existing chain with free 3' OH group at one end

- (d) Describe **two** ways in which the functions of molecules **P** and **Q** are

- (i) similar to each other; [2]

Type of bond formed: phosphoester bond between nucleotides

Template used: DNA as template

Direction of reading the template: from 3' to 5' direction

- (ii) different from each other. [2]

Feature	Replication	Transcription
Start site	Begins at regions of DNA known as <u>origins of replication</u>	Begins at regions of DNA known as <u>promoter</u>
Nucleotides base pair with template	Free deoxyribonucleotides/ deoxyribonucleoside triphosphate	Free ribonucleotides/ ribonucleoside triphosphate
Base pairing rule	A=T and C≡G	A=U and C≡G

[Total: 12]

5. In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Eugene and Cornelius crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Eugene's and Cornelius's crosses were recorded and are shown in **Table 5.1**.

Table 5.1

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Eugene's cross	23	4	3	20
Cornelius's cross	3	21	23	3

- (a) Describe briefly how a plant breeder ensures that the offspring produced are **only** from the desired cross. [2]

- anthers removed (before maturity) (to produce male sterility)
 - pollen transferred by hand
 - plants isolated / flowers bagged (before and after pollination)
- any two

- (b) Using the results from Eugene's cross,

- (i) Carry out a χ^2 test using the formula below.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = \frac{(23-12.5)^2}{12.5} + \frac{(4-12.5)^2}{12.5} + \frac{(3-12.5)^2}{12.5} + \frac{(20-12.5)^2}{12.5};$$

$$= 8.82 + 5.78 + 7.22 + 4.50 = 26.32;$$

[1]

5. (b) (ii) With reference to the table of probabilities provided, explain how the value for the χ^2 test supports the hypothesis that the two genes are independently segregating.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

[2]

1. Given $\chi^2 = 26.32$, $v=3$, the probability of the deviation is due to chance is < 0.001 which is less than 0.05
2. Hence, reject H_0 in favour of H_A
3. There is significant difference between the observed and the expected results
4. The two genes are not independently segregating

- (c) Using the symbols given, state the genotypes of the tomato plants in Eugene's and Cornelius's crosses.

[2]

Eugene's cross: $\frac{A B}{a b} \times \frac{a b}{a b}$

Cornelius's cross: $\frac{A b}{a B} \times \frac{a b}{a b}$

Reject: different symbols used

::

::

(d) Draw a genetic diagram to explain the results of Cornelius's cross. [5]

Parental phenotype:	Red fruit with green base;	x	Orange fruit with no green base;
Parental genotype:	$\frac{A b}{a B}$		$\frac{a b}{a b}$
Gametes:	$\frac{A b}{a B}$ $\frac{A B}{a b}$;		$\frac{a b}{a b}$;
Punnet square	Parental gametes		Recombinant gametes
	$\frac{A b}{a b}$	$\frac{a B}{a b}$	$\frac{A B}{a b}$ $\frac{a b}{a b}$
$\frac{a b}{a b}$	$\frac{A b}{a b}$	$\frac{a B}{a b}$	$\frac{A B}{a b}$ $\frac{a b}{a b}$
	Red fruit with no green base	Orange fruit with green base	Red fruit with green base Orange fruit with no green base
	21	23	3 3 ;
	Parental		Recombinants ;

Correct genotype;;
Correct phenotype;;

[Total: 12]

6.

PART I

Fig. 6.1 is a diagram of an electron micrograph of part of a chloroplast showing thylakoid membranes.

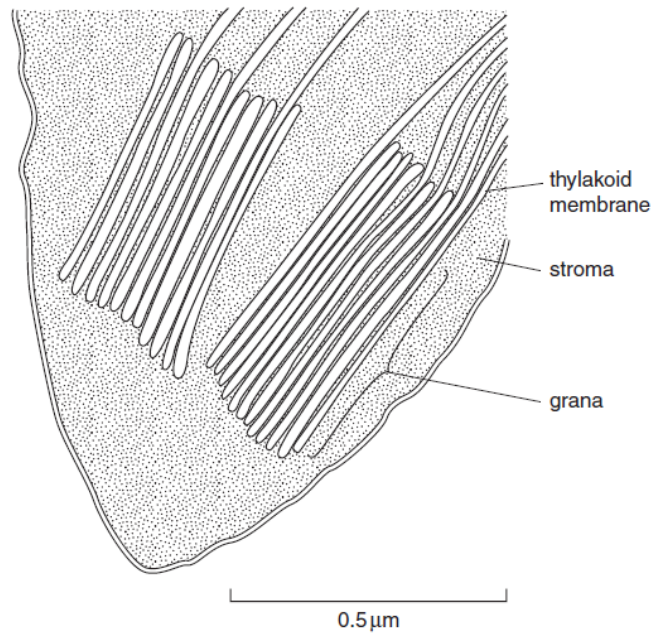


Fig. 6.1

(a) Describe the role of thylakoid membrane in photosynthesis.

[4]

1. Large surface area for embedment of
2. photosynthetic pigments for harvesting light energy
3. ATP synthase for phosphorylation of ADP to ATP
4. photosystems to generate ATP & NADPH (for Calvin cycle)
5. electron transport chain for photophosphorylation
6. Impermeable to protons
7. allow for build-up of proton gradient for chemiosmosis

6.

PART II

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to NADP⁺.

Fig. 6.2 shows the root hairs in the two variant of *A. thaliana*.

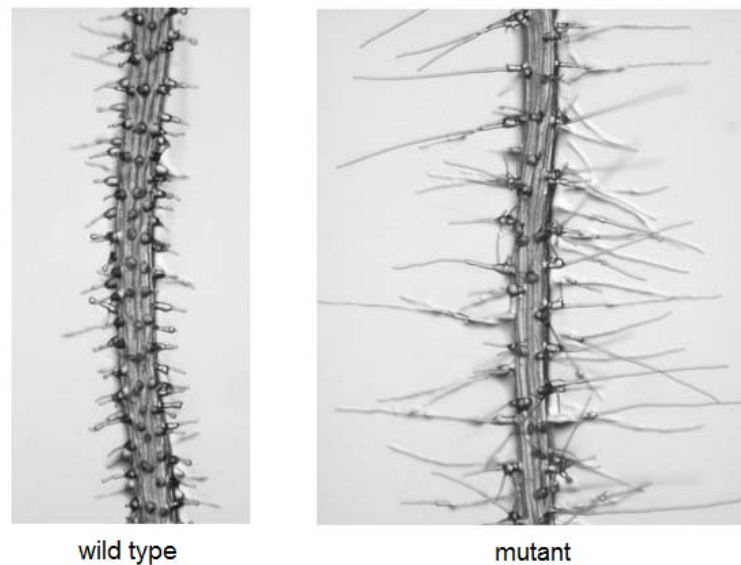


Fig. 6.2

(b) Explain the role of NADPH in photosynthesis.

[2]

1. Provides reducing power in Calvin cycle
2. to convert 1,3-bisphosphoglycerate to glyceraldehyde-3-phosphate

6. In a separate experiment, the length of root hairs in wild type and mutant *A. thaliana* were measured at intervals, together with the activity of NADP oxidase in the tips of the root hairs. The results are shown in **Fig. 6.3**.

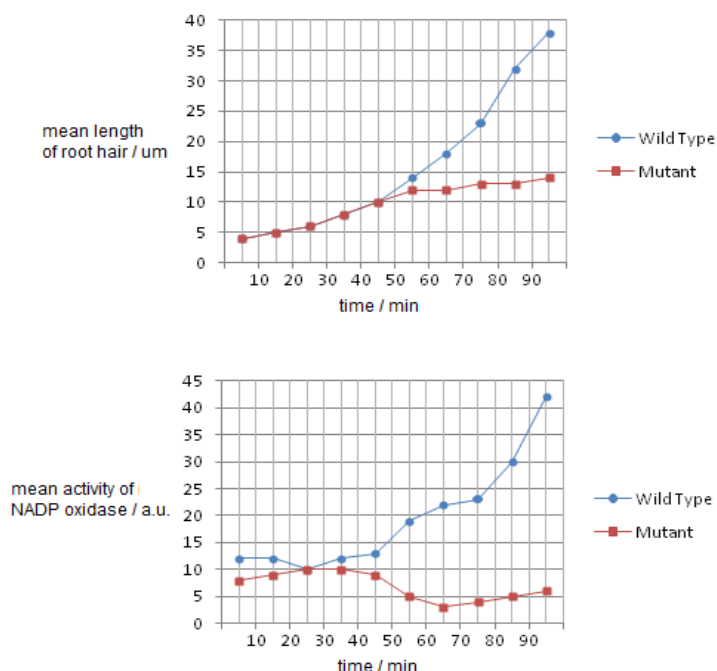


Fig. 6.3

- (c) With reference to **Fig. 6.3**, and your knowledge of photosynthesis, describe and explain the difference in the growth of root hairs in the two types of *A. thaliana*.

[4]

Describe

45 to 95 min,

1. Mean length of root hair for wild type *A. thaliana* increases from 10um to 38um
2. while that for mutant *A. thaliana* increases from 10um to 14um
3. Mean activity of NADP in wild type *A. thaliana* increases from 13au to 42au
4. while that in mutant *A. thaliana* decreases from 9au to 6au

Explain

5. NADP oxidase catalyses the conversion of NADPH to NADP⁺ for light reaction of photosynthesis

Accept: Calvin cycle of photosynthesis

6. where ATP is synthesised; for cell growth / growth of root hair

Accept: synthesis of glucose (if Calvin cycle mentioned in point 5)

[Total: 10]

7. **Fig. 7.1** shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates.

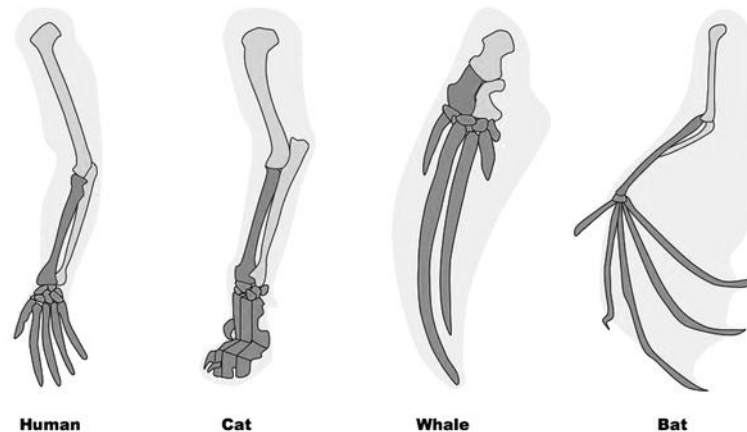


Fig. 7.1

- (a) Explain how the relationship between the structures in **Fig. 7.1** provides evidence to support the theory of evolution. [2]
1. Same arrangement of bones in the pentadactyl forelimbs of four vertebrates
 2. Though they have different functions
 3. Therefore they were likely to have originated from a common ancestor
 4. Had been subject to different selection pressures
- (b) Explain how fossil evidence can also be used as evidence to support the idea that evolution has taken place? [2]
1. fossils show changes over time
Accept: many fossil organisms dissimilar from modern organisms / general trend from, small / simple, to, large / complex
 2. fossils or rocks can be dated
Accept: idea of fossils in chronological order
 3. fossils showing intermediate forms
 4. many fossils no longer exists

7. (c) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing.

[3]

1. Variation in the skeletal structure
2. due to mutations/ presence of different alleles
3. Those with favourable phenotype (have a selective advantage) survive to reproduce
4. Pass on the allele to the offspring
5. Those at selective disadvantage will not survive
6. An increase in frequency of the allele for the favourable phenotype

7. The pipistrelle is the most common species of bat in Europe. **Fig. 7.2** shows a map of Canary Islands, a Spanish archipelago located just off the northwest coast of mainland Africa. El Hierro is the most recent Canary Island, having appeared approximately 1.2 million years ago.

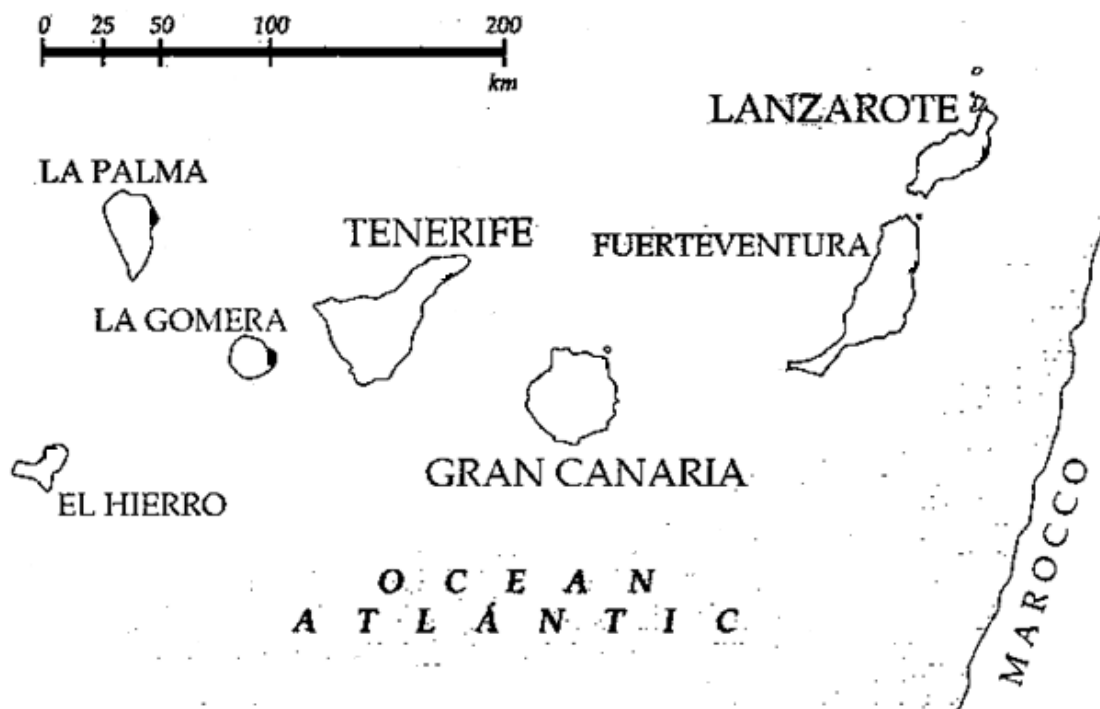


Fig. 7.2

7.

It was originally thought that all pipistrelles belonged to the same species. However, mtDNA reveals that there were two species: *Pipistrellus maderensis* and *Pipistrellus kuhlii*. The data for both species are provided in **Table 7.1**.

Table 7.1

Species	Mean body mass (g)	Mean wingspan (cm)	Colour
<i>P. maderensis</i>	5.7	3.0	dark brown fur with light wing markings
<i>P. kuhlii</i>	5.5	3.0	light brown to reddish brown with light wing markings

(d) Using the data in **Table 7.1**, suggest why pipistrelle were originally classified as one species. [1]

1. similar, (body) mass, i.e. 5.5 g
2. same wingspan, i.e. 3.0 cm
3. same wing markings, i.e. light wing marking
4. previously unable to measure sequence similarity

Evolution of the two bat species, *Pipistrellus maderensis*, *Pipistrellus kuhlii*, was studied by comparing ~1 kbp of mtDNA (from *cytochrome b* and 16S rRNA genes) from the bats in different islands.

Sequence divergence of *P. maderensis* is the lowest between La Gomera and El Hierro as compared to between La Gomera and the remaining islands (i.e. Tenerife and La Palma).

(e) Using the above information and **Fig. 7.2**,

(i) explain why the sequence divergence of *P. maderensis* is the lowest between La Gomera and El Hierro. [3]

1. a recent colonization of El Hierro by *P. maderensis* from La Gomera
2. as the island was formed only 1.2 mya, insufficient time for evolution
3. similar selection pressure

- (ii) suggest why no gene flow occurs between the *P. maderensis* on La Gomera and El Hierro. [1]

The distance between La Gomera and El Hierro is greater than between other islands

[Total: 12]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. (a) Describe the structure of the cell membrane and relate it to its properties [7]

Structure	Property – Selective permeability
1. comprises of phospholipid bilayers 2. hydrophobic tails facing inwards and hydrophilic heads facing the aqueous medium 3. hydrophobic barrier	P1. permeable to non-polar / hydrophobic substances
S2a. interspersed with proteins S2b. integral proteins are embedded in the membrane and S2c. peripheral proteins are not embedded in the membrane	P2. transmembrane proteins allow transport of ions/ polar/ hydrophilic substances across the membrane
Structure	Property – Fluidity
S3. phospholipids can move laterally	P3a. fluidity P3b. can reseal itself after it is disrupted P3c. can invaginate and form vesicles P3d. transient gap for movement of small, uncharged &/ non-polar molecules
S4. cholesterol wedged between phospholipid molecules	P5. regulates the fluidity
Structure	Property – Cellular identity
S5a. presence of glycolipids and glycoproteins; S5b. carbohydrate chains face the external matrix outside the cell	P5a. membrane is asymmetrical P5b. carbohydrate chain serves as cell surface marker / cellular recognition

(b) Explain the significance of mitosis in growth, repair and asexual reproduction.

[4]

1. Daughter cells are genetically identical to parent cell
2. contain the same number of chromosomes as the parent cell
3. Due to semi-conservative replication of DNA during interphase
4. results in genetic stability within populations of cells derived from the same parental cells
5. daughter cells can then function harmoniously as part of the same tissue, organ and organism

Growth

6. allows the growth and development of a zygote into a multicellular organism
7. forming new cells that are genetically identical to the existing cell(s)

Repair

8. produces new cells required to repair worn-out or damaged tissues
9. to return a tissue to its former condition

Asexual reproduction

10. to produce genetically identical daughter cells that can form new offspring
11. to reproduce rapidly in an environment that the parent plant is well adapted to
12. colonises the habitat much more quickly than via sexual reproduction
13. AVP

- (c) Outline the behaviour of chromosomes during meiosis I and explain the significance of each phase in leading to genetic variation. [9]

Phase	Event	Significance
Prophase I	1. Chromatin <u>condenses</u> to form chromosome 2. Chromatin forms looped domains around nonhistone protein 3. further coiling to form chromosomes 4. each chromosome consists of 2 sister chromatids joined at centromere 5. DNA replicated to form 2 daughter DNA during interphase 6. Homologous chromosomes pair up to form bivalent - synapsis 7. Chiasma formed 8. Crossing over between non-sister chromatids of homologous chromosomes 9. At the end of prophase, each member of homologous pair of chromosome is attached to spindle fibres at the centromere	S1. New combination of alleles
Metaphase I	10. Homologous chromosomes align at metaphase plate 11. Orientation of each bivalent is <u>independent</u> of other bivalents	S2. Random separation of homologous chromosomes
Anaphase I	12. Homologous chromosomes separate 13. pulled to opposite poles of cell 14. centromere leading	S3. Halving of chromosome number S4. Results in variation in combination of paternal and maternal chromosomes in gametes
Telophase I	15. Chromosome arrive at opposite poles of the cell 16. Chromosomes decondense to form chromatin	S5. Haploid set of chromosomes / forming 2 haploid cells for fertilisation

[Total: 20]

9. (a) Discuss, using named examples, how dual control of gene expression is achieved in *Escherichia coli*.

[8]

Negative gene regulation: Scenario 1 or 2 [max 4 marks]

Scenario 1

1. Negative gene regulation
2. by lac repressor protein
3. Lactose present
4. Lactose enter cell due to a few molecules of permease
5. Lactose is converted to allolactose by β -galactosidase
6. Allolactose binds to allosteric site on lac repressor protein
7. causing lac repressor protein to adopt its inactive conformation
8. Lac operon switched on

Scenario 2

9. Negative gene regulation
10. by lac repressor protein
11. Lactose absent
12. thus no allolactose bound to lac repressor protein
13. allowing for synthesis of small amounts of lac repressor protein
14. Lac repressor protein is in its active conformation
15. Lac repressor protein binds to the operator
16. Lac operon switched off

Positive gene regulation: Scenario 3 or 4 [max 4 marks]

Scenario 3

17. Positive gene regulation
18. by catabolite activator protein
19. Glucose absent, lactose present
20. Level of cAMP is high
21. Level of CAP-cAMP complex is high / CAP assumes active conformation
22. CAP-cAMP complex binds to promoter
23. increases affinity of RNA polymerase for promoter
24. stimulate / increases rate of transcription of structural genes

Scenario 4

25. Positive gene regulation
26. by catabolite activator protein
27. Glucose present, lactose present
28. Level of cAMP is low
29. Level of CAP-cAMP complex is low / most CAP assumes inactive conformation
30. CAP-cAMP complex does not bind to / detaches from promoter
31. Low affinity of RNA polymerase for promoter
32. Low rate of transcription of structural genes

(b) Explain the significance of operons in bacteria.

[4]

1. Operons in bacteria group together genes
2. that are involved in a common metabolic pathway
3. for coordinated control of gene expression
4. by a single promoter and operator in that operon
5. This allows for the structural genes in an operon to be transcribed together
6. resulting in the concurrent synthesis of proteins involved in the same pathway
7. Operons enable bacteria to only synthesize the required proteins
8. This preventing wastage of energy and resources / allocating resources and energy for other use
9. Operons also allow for bacteria to use a variety of sugars
10. and preferentially use resources to gain selective advantage

(c) Explain how the structure of neurone facilitates impulse transmission.

[8]

	Structure	Function
1.	Dendrites	♦ Extensively branched so that many synapses can be made with many other neurons
2.	Numerous mitochondria in cell body and axon	♦ Produce ATP for sodium-potassium pumps / transport of vesicles / synthesis of proteins
3.	Axon	♦ Conducts impulses away from cell body
4.	Nissl's granules	♦ Synthesis of proteins such as Na ⁺ -K ⁺ pump / acetylcholinesterase
5.	Myelin sheath	♦ Acts as an electrical insulator, preventing the formation of action potentials
6.	Nodes of Ranvier	♦ Enable action potential to jump from node to node, increasing the speed of transmission of a nerve impulse
7.	Presence of synaptic vesicles in synaptic knob	♦ Synaptic vesicles contain neurotransmitters for synaptic transmission
8.	Na ⁺ -K ⁺ pump	♦ Pumps 3 Na ⁺ out of in exchange for 2 K ⁺ into axoplasm for maintenance of resting potential
9.	Voltage-gated Na ⁺ / K ⁺ / Ca ²⁺ channel	♦ Allows for influx of Na ⁺ / efflux of K ⁺ / efflux of Ca ²⁺ upon trigger by change in membrane voltage
10.	Chemically-gated Na ⁺ channel	♦ Allows for diffusion of Na ⁺ upon binding of neurotransmitter
11.	AVP	

[Total: 20]